CRA
Computing Research Association

NAIRR Pilot

Classroom Expansion Conferences

- Research Emerging - Large

Building AI Research Pathways from Undergraduate to Doctoral Programs

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Why This Matters: The State of AI in Higher Education

86%

of employers anticipate AI will drive business transformation in 5 years

WEF Future of Jobs Report, 2025

3%

of employers believe higher education is adequately preparing AI-ready graduates

Digital Education Council, AI in the Workplace, 2025

6,000+

students supported by NAIRR across all 50 U.S. states since 2024

NSF NAIRR 2-Year Progress Update, 2025

Workforce Demand

~39% of core job skills will change by 2030. AI & big data are the fastest-growing category, with nearly 9 in 10 employers rating their importance as rising.

WEF Future of Jobs Report, 2025

Curricula Not Keeping Up

AI has not been broadly integrated into higher-education curriculum, instruction, and assessment, and faculty-facing adoption remains uneven and under-examined.

Liang et al., 2025

AI Improves Learning Outcomes

AI integration in higher education raised average grades to 88% and increased retention rates to 85%.

Computer Applications in Engineering Education, 2025

University of the Cumberlands — AI Degree Pathway

One of a limited number of U.S. institutions offering a clear, seamless B.S. → M.S. → Ph.D. progression in Artificial Intelligence

UNDERGRADUATE

B.S. in Artificial Intelligence

60 Credit Hours* | 42 Core AI Hours

100% Online | Launched 2026

- Machine learning & deep learning
- Natural language processing (NLP)
- AI ethics & responsible use
- Hands-on projects with real datasets
- Pathway to M.S. & Ph.D. in AI

First fully online Bachelor degree in Kentucky

Note: Credits do not include Gen Ed courses.

→

GRADUATE

M.S. in Artificial Intelligence

31 Credit Hours

Online & Executive Hybrid |

- Neural networks & robotics
- Data mining & information security
- Business intelligence & IT management
- Ethical AI practices

Program anchor — launched 2022

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DOCTORAL

Ph.D. in Artificial Intelligence

60 Credit Hours

Online & Executive Hybrid |

- Advanced ML, deep learning, reinforcement learning
- Data engineering & NLP
- End-to-end projects & research assignments

Coursework + dissertation

NAIRR Connection & Implications

NAIRR Compute Access + **Developing AI-Ready Graduates**

= AI Workforce Readiness at Scale

- > NAIRR Pilot expands compute access — This model empowers us to build the human infrastructure needed to create meaningful connections.
- > Advances NAIRR objectives at institutions with large, diverse computing enrollments
- > Broadens participation in AI research among historically underserved populations

Source: Wang & Muro, 2024; NSF DCL 24-109, 2024

Vertically Aligned Curricula

A practical framework for emerging institutions, focusing on foundational literacy and research training.

Scalable Broadening of Participation

Because smaller institutions struggle with limited AI expertise and computing resources, a scalable, NAIRR-enabled model is key to broadening participation among diverse student populations that have traditionally faced challenges in research opportunities.

Key Takeaways

AI Is Here to Stay

AI is not going anywhere. Higher education must find better ways to prepare students — and that means offering AI degree programs across the board, at every level and in every discipline.

NAIRR Levels the Playing Field

NAIRR is vital — providing AI resources not only to large computing institutions, but to the small and medium institutions that lack access. Equitable infrastructure is how we broaden participation.

References

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